INCH-POUND
A-A-59507
30 May 2001
SUPERSEDING
MIL-C-52286/22
4 May 1970

COMMERCIAL ITEM DESCRIPTION

CABLE ASSEMBLY, POWER ELECTRICAL, ADAPTER

The General Service Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers a cable assembly adapter used for interior and exterior power and lighting (see figure 1). The adapter has molded male and female plugs, and pins (see figures 2 through 4).

2. SALIENT CHARACTERISTICS

2.1 Jacketing material. The jacketing material shall be medium-low temperature, heat resistant with the flexibility and durability of styrene-butadiene rubber (SBR) in accordance with the American Society for Testing and Materials (ASTM) D 518, "Standard Test Method for Rubber Deterioration - Surface Cracking" (DoD adopted).

2.1.1 Temperature range. The ambient temperature range shall be -55 ºC for the jacket material to +75 ºC for the conductors in accordance with ASTM D 1053, "Standard Test Methods for Rubber Property - Stiffening at Low Temperatures: Flexible Polymers and Coated Fabrics" (DoD adopted).

2.1.2 Overheating. Applying 600 volts (V) at 15 amperes (A) shall cause no overheating, arcing, or damage in accordance with ASTM D 1149, "Standard Test Method for Rubber Deterioration - Surface Ozone Cracking in a Chamber" (DoD adopted).

2.1.3 Hardness. The Shore Durometer "A" hardness shall be 60 +/- 10.

2.1.4 Insulation. The jacket material shall conform to the requirements of ASTM D 1765, "Standard Classification System for Carbon Blacks Used in Rubber Products" (DoD adopted).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 6150

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2.2 Pin size. The end of pin hemispherical radius shall be one-half the pin diameter. The male pin shall be 0.594 to 0.656 high, and 0.154 to 0.156 inches wide as shown in figure 4.

2.3 Socket rounding. Female sockets shall be rounded for centering with metal bands on sockets. The socket depth for connector "A" shall be 0.594 inches to 0.656 inches. The socket width for connector "A" shall be 0.154 to 0.156 inches. The socket depth for connector "B" shall be 0.641 to 0.703 inches. The socket width for connector "B" shall be 0.156 to 0.158 inches

2.4 Connector strength. The minimum bond between connector and cable shall be 150 pounds (lbs.) (see figure 1).

2.5 Connect/disconnect force. The connect/disconnect force shall be from 10-35 lbs.

2.6 Dielectric strength. The molded components shall withstand five times the rated voltage (600 V at 15 A) for one minute without damage, arcing, or breakdown.

2.7 Voltage drop. The voltage drop shall not be more than 10 mV at 15 A.

2.8 Adapter storage requirements. The adapter shall not be damaged when stored at -65 to 125 ºF at maximum humidity in accordance with ASTM D 518, ASTM D 1053, and ASTM D 1149.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, and conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

4.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

4.2.1 The manufacturer/distributor of the item must have been producing a product meeting the requirements of this CID for at least 3 years.

4.2.2 The company must provide a warranty for replacing defective items.

5. PACKAGING. Preservation, packing, and marking, shall be specified in the contract or order (see 6.3(b)).

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6. NOTES

6.1 Part or identification number (PIN). The PIN procedure is AA59507-1. It is for government purposes and does not constitute a requirement for the contractor.

6.2 Source of documents.

6.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

6.2.2 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.3 Ordering data. Acquisition documents must specify the following:

a. CID document number, revision, and CID PIN.
b. Packaging requirements (see 5).

6.4 National stock number (NSN). This CID is associated with NSN 6150-00-549-5401. Other NSNs may also correspond with this document.

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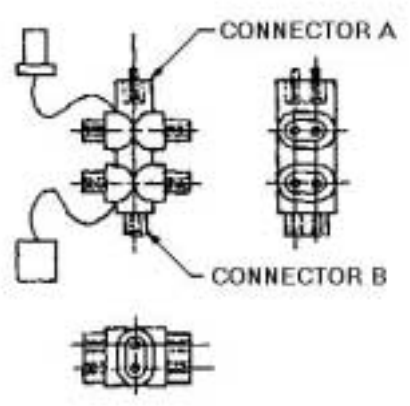

FIGURE 1. Cable assembly, adapter.

Notes:
1. Connector A - see figure 2
2. Connector B - see figure 3
3. Minimum bond between connector and cable = 150 lbs.

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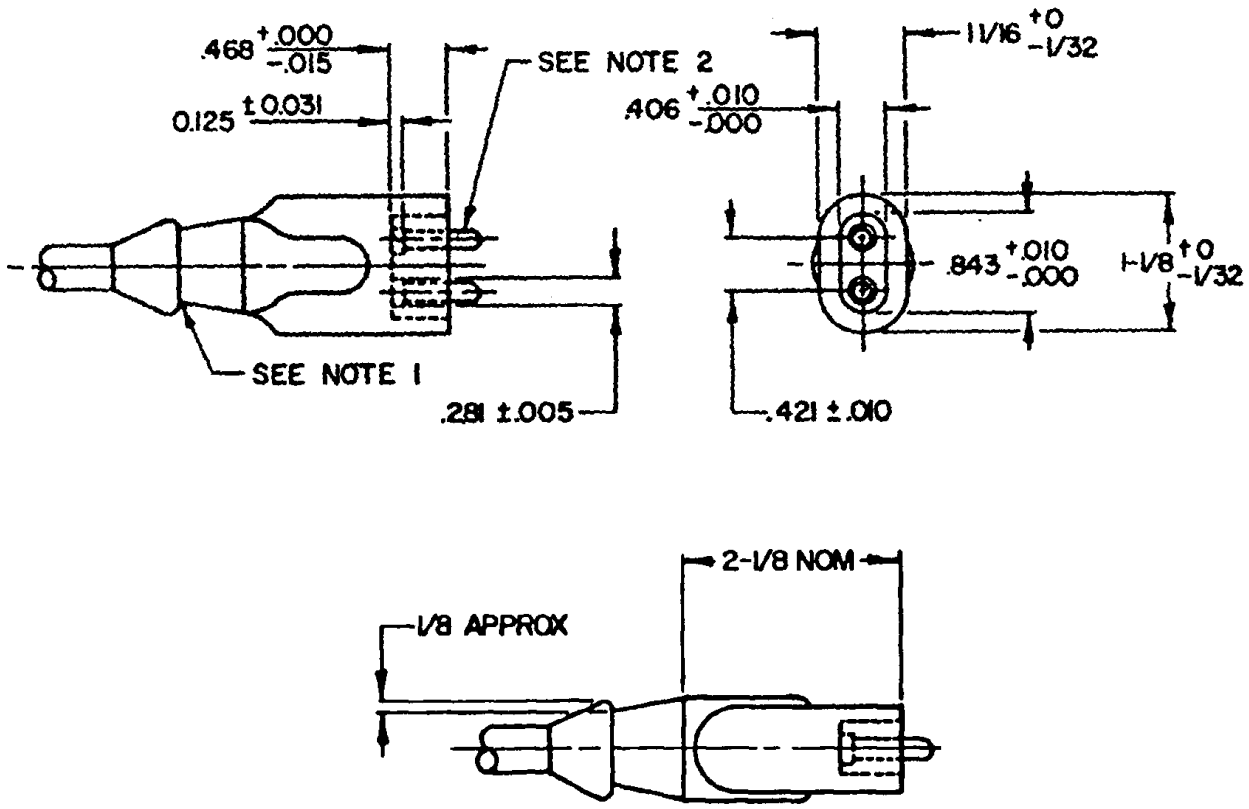

FIGURE 2. Molded plug, male.

Notes:

1. The groove shall be of any shape with sufficient 0.010 inch = 0.0254 cm depth, width, or cross section to support a 1/8 inch 0.015 inch = 0.0381 cm twine or cord.
2. Pin dimensions (see figure 4)
   A = 0.594 - 0.656
   B = 0.154 - 0.156
3. Connector rating: 600 V at 15 A
4. Voltage drop: Not more than 10 mV at 15 A
5. Connect/disconnect force = 10 - 35 lbs

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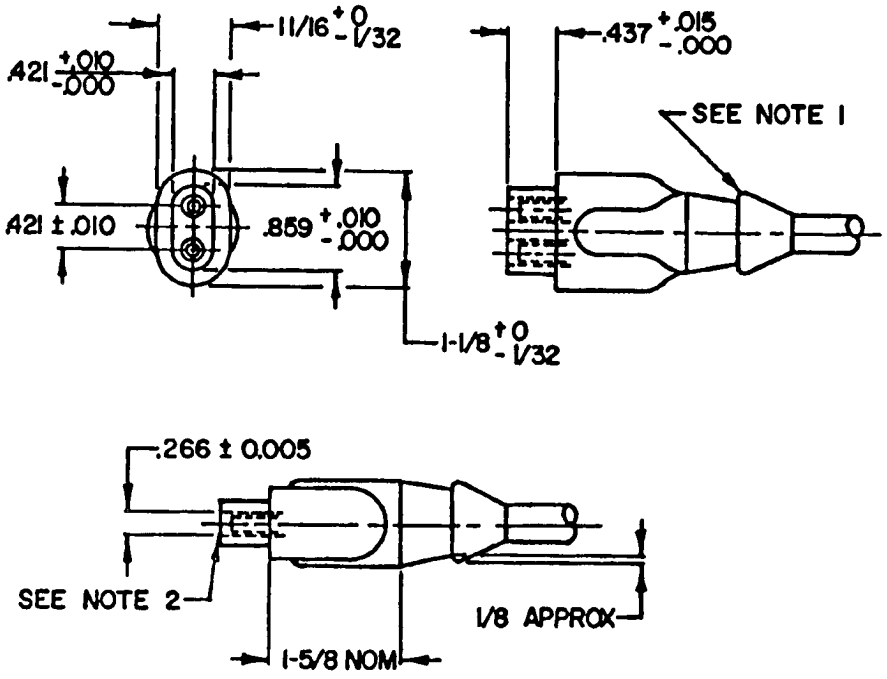

FIGURE 3. Molded plug, female.

Notes:

1. This groove shall be of any shape with sufficient depth, width, or cross section to support 1/8 inch twine or cord.
2. Female contact dimensions (see figure 4):
   C = 0.641 - 0.703
   D = 0.156 - 0.158
3. Contacts shall be recessed by 0.125 inch (+/- 0.0156 inch) below face.
4. Connector rating: 600 V at 15 A
5. Voltage drop: not more than 10 mV at 15 A
6. Connect/disconnect force = 10 - 35 lbs

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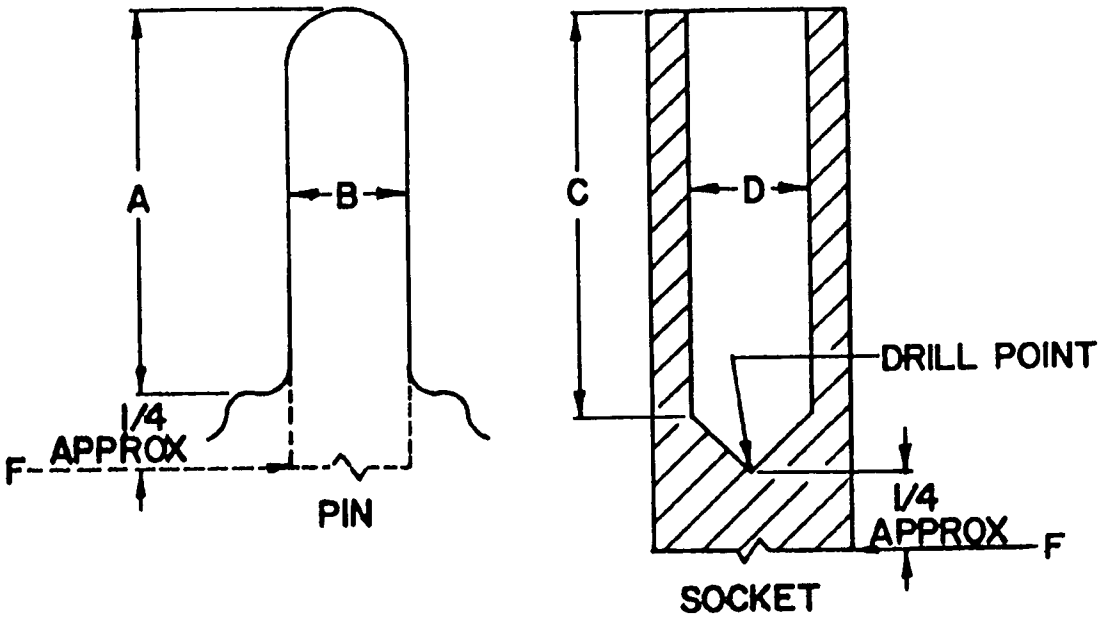

FIGURE 4. Pin configuration.

Note:
1. F - Point for voltage drop on contact.

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MILITARY INTERESTS:

Custodian:
Army - CR4

Reviewer:
Army - AT

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FXE

Preparing activity:
DLA - GS2

(Project 6150-0282)

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